CLINICAL INDICATION: A 72-year-old female presenting with progressive New York Heart Association (NYHA) functional class III dyspnea on exertion, orthopnea, and bilateral lower extremity pitting edema.

2-D AND DOPPLER ECHOCARDIOGRAPHIC MEASUREMENTS:

Left Ventricle: Left ventricular internal dimension at end-diastole is 5.4 cm (normal 3.8-5.2 cm). Left ventricular internal dimension at end-systole is 3.9 cm. Interventricular septal thickness in diastole is 1.3 cm; posterior wall thickness is 1.2 cm, consistent with moderate concentric left ventricular hypertrophy. Left ventricular ejection fraction (LVEF) quantitatively estimated at 38% by modified Simpson's biplane method. Qualitative evaluation demonstrates regional wall motion abnormalities characterized by hypokinesis of the anterior, anteroseptal, and apical myocardial segments.

Diastolic Function: Mitral inflow Doppler demonstrates E/A ratio of 2.1 with deceleration time of 135 ms. Tissue Doppler septal e-prime velocity is 4.8 cm/s, yielding an elevated average E/e' ratio of 18.4, consistent with grade III restrictive left ventricular diastolic filling pattern and severely elevated left atrial filling pressures.

Right Ventricle: Right ventricle is mildly dilated with basal diameter of 4.2 cm. Tricuspid annular plane systolic excursion (TAPSE) measures 1.4 cm, indicating mild-to-moderate right ventricular longitudinal systolic dysfunction.

Valvular Assessment: The aortic valve is trileaflet, fibrocalcific, with restricted systolic excursion; peak transaortic velocity is 2.8 m/s with mean systolic gradient of 16 mmHg (consistent with moderate aortic sclerosis/mild stenosis). The mitral valve leaflets demonstrate myxomatous thickening with moderate-to-severe posteriorly directed eccentric regurgitant jet (regurgitant volume 48 mL, effective regurgitant orifice area 0.32 cm2). Trace tricuspid regurgitation is present with peak velocity of 3.4 m/s; utilizing an estimated right atrial pressure of 15 mmHg based on a dilated inferior vena cava measuring 2.4 cm with <50% inspiratory collapse, the estimated pulmonary artery systolic pressure (PASP) is elevated at 61 mmHg, reflecting moderate-to-severe pulmonary arterial hypertension.

Pericardium: No pericardial effusion is visualized.

CONCLUSION:

1. Moderate concentric left ventricular hypertrophy with depressed global systolic function (LVEF 38%).
2. Hypokinesis of anterior and apical segments.
3. Grade III restrictive diastolic dysfunction with high left ventricular filling pressures.
4. Moderate-to-severe eccentric mitral regurgitation.
5. Moderate pulmonary hypertension (estimated systolic PAP 61 mmHg).